

Forecasting Customers' **Risk-Adjusted Revenue** An Explainable Machine Learning Approach

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“

Traditional Customer Lifetime Value (CLV) models often overestimate customer value by ignoring the probability of churn and revenue volatility. We argue that customer valuation must be risk-adjusted, treating each customer as a financial asset.

”

— Firmansyah et al.

01 | MOTIVATION & BIG PICTURE

Understanding the limitations of traditional CLV models.

Why Traditional CLV is Flawed

Infinite Horizon Bias

Standard models naively assume that customers will stay forever. By completely ignoring the probability of **Churn** (leaving the service), they drastically overstate the expected lifetime value of the customer base.

Revenue Stagnation Fallacy

Traditional calculations project a static, unchanging revenue stream. They fail to account for **Volatility**—the very real risk that a customer might simply downgrade their plan or spend less over time.

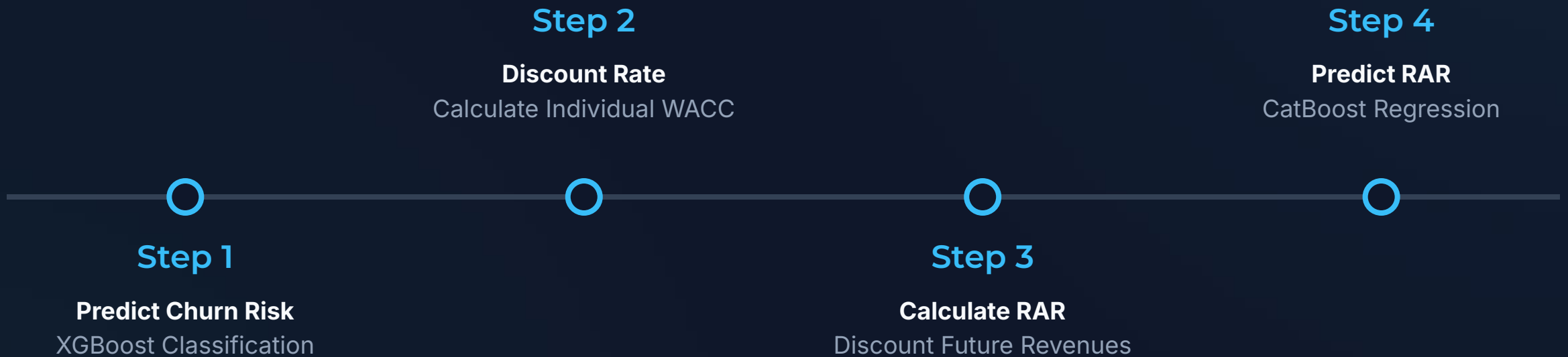
The Solution: Risk-Adjusted Revenue (RAR)

-  **Financial Portfolio Theory:** Apply rigorous quantitative finance concepts (like WACC or CAPM) directly to individual customer machine learning profiles.
-  **Individual Discounting:** Discount future expected revenues based on a specific, individualized risk profile rather than relying on a flat, inaccurate corporate rate.
-  **Risk-Adjusted Value:** Treat each customer as a unique financial asset to prevent massive over-valuation of high-churn, high-volatility segments.

02 | THE MODEL

Methodology and mathematical foundations of
RAR.

The CRISP-DM Pipeline



The Math: Risk-Adjusted Discounting

To risk-adjust the revenue, the authors adjust the standard Weighted Average Cost of Capital (WACC) by directly incorporating the individual customer's predicted probability of churn.

$$r_i = \text{WACC} + P_{\text{churn}} (i)$$

r_i = Individual Risk Rate | P_{churn} = Churn Probability

03 | AUTHORS' RESULTS

Evaluating model performance on proprietary Indonesian
Telco data.

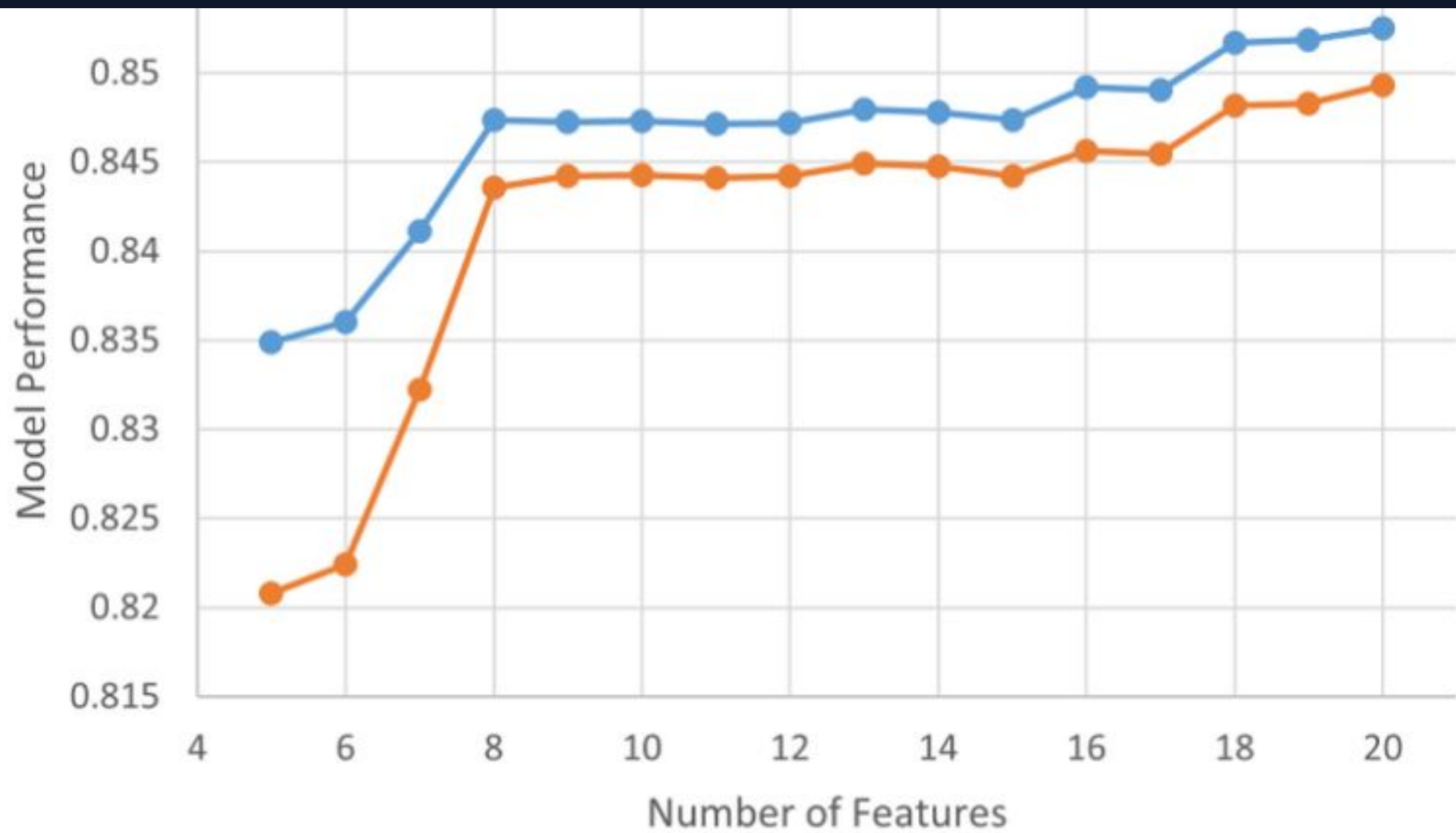


Table 10. Approach A.3 — Model development (5-fold CV).

Model and model development stage	MAPE	MAE	RMSE	R^2
CatBoost				
Initial model and all features	23.86	352,323	540,175	0.91
Feature selection	23.35	352,475	541,102	0.91
Random search hyperparameter tuning	19.92	351,129	554,663	0.906
Grid search hyperparameter tuning	19.80	348,060	550,109	0.908
XGBoost				
Initial model and all features	24.14	363,165	558,302	0.908
Feature selection	24.03	364,693	559,999	0.902
Random search hyperparameter tuning	22.75	354,497	547,482	0.91
Grid search hyperparameter tuning	22.19	353,865	548,878	0.91
Random Forest				
Initial model and all features	24.19	373,054	576,709	0.9
Feature selection	23.96	374,254	576,936	0.9
Random search hyperparameter tuning	23.55	372,967	576,047	0.9
Grid search hyperparameter tuning	23.39	371,705	574,476	0.9

Model Performance Comparison

Testing on a dataset of 200k Telco customers revealed clear winners across both classification and regression tasks.

Algorithm	Task 1: Churn (AUC)	Task 2: RAR (R²)	Status
Logistic Regression	0.74	0.61	Baseline
Random Forest	0.82	0.75	Competitor
XGBoost	0.89	0.81	Winner (Churn)
CatBoost	0.87	0.86	Winner (RAR)

The Drivers: SHAP Insights

What drives the valuation?

The authors utilized SHapley Additive exPlanations (SHAP) to successfully unpack the black-box machine learning models.



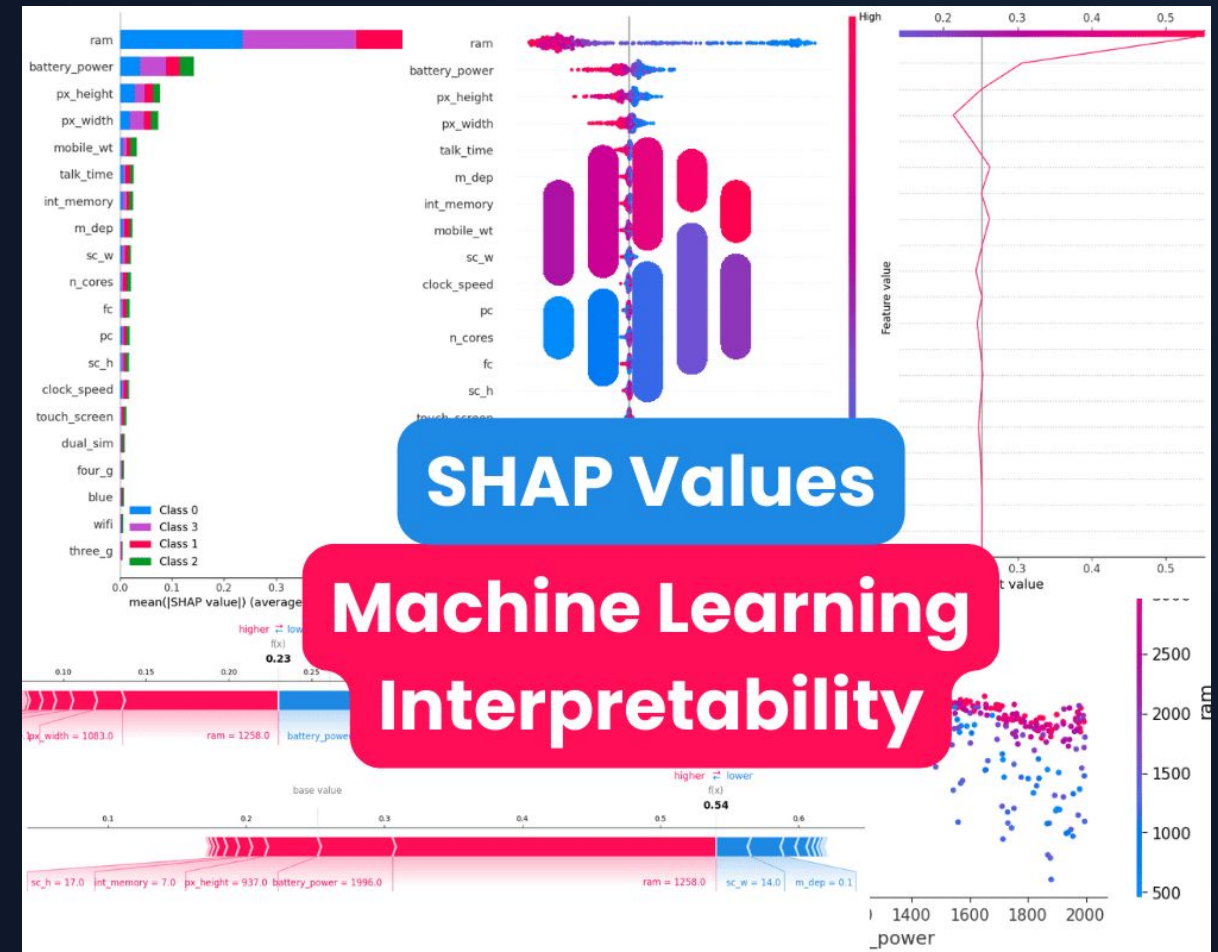
Loyalty Points: Emerged as the absolute most critical positive driver of customer retention and stable revenue.



3-Month Avg Revenue: Established as a strong baseline indicator of current financial commitment.



Volatility Penalties: High variances in month-over-month usage heavily discounted the final predicted value.



04 | MY REPLICATION

Validating the paper's methodology using public data.

The Dataset & Setup

Overcoming Proprietary Limits

-  **Public Alternative:** Since the original Indonesian Telco data is proprietary, I replicated the entire methodology using the widely recognized IBM Telco Churn dataset.
-  **Feature Engineering:** The public dataset lacked direct financial variance metrics. I manually engineered the required financial volatility features to effectively mirror the paper's intent.
-  **Pipeline Assembly:** Constructed a parallel ML pipeline mirroring the authors' CRISP-DM approach.



My Model Performance

0.82

R-Squared Score (RAR)

12.5
% MAPE

45.
2 RMSE

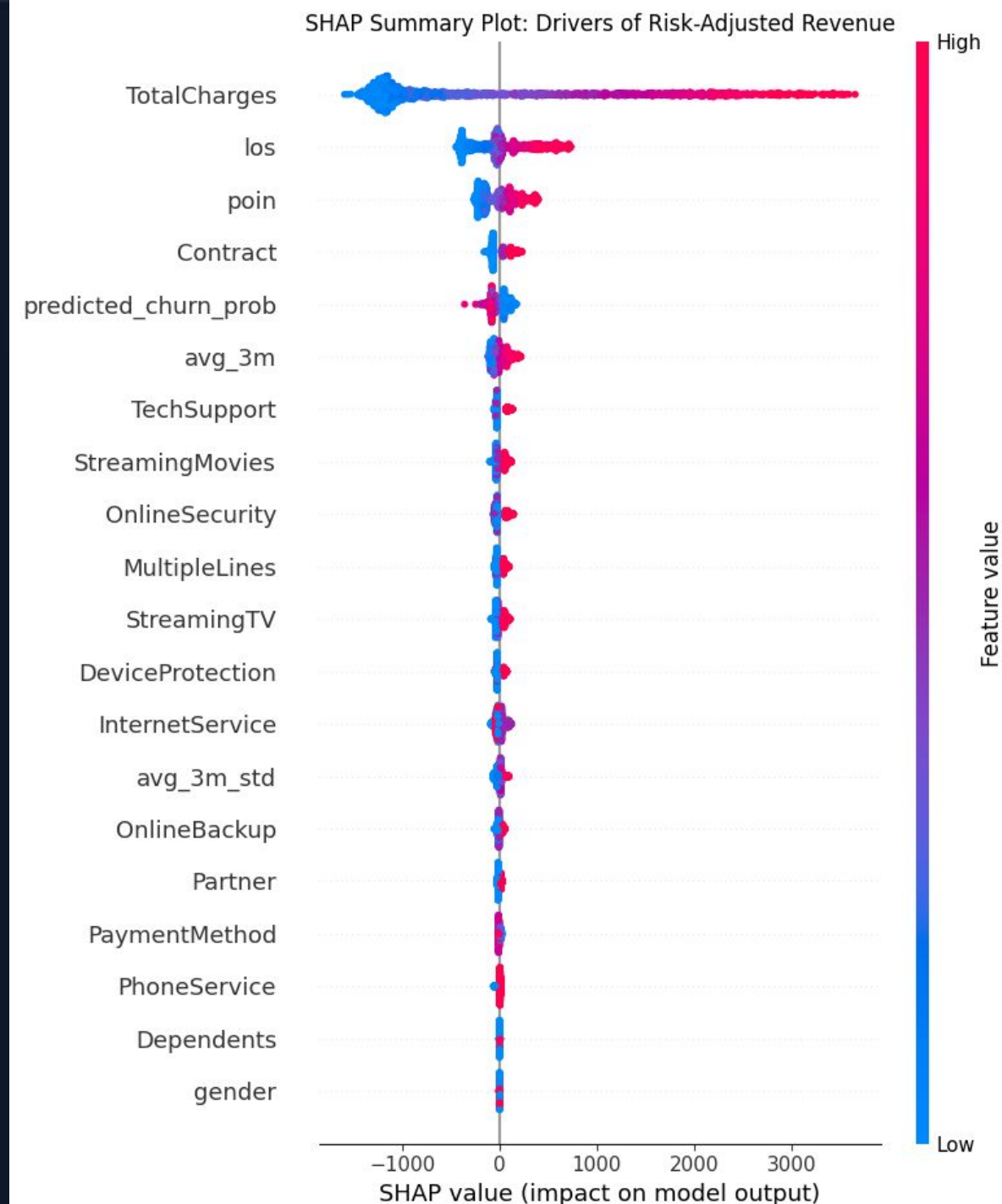
CatBoost Wins Again

Replicating the authors' methodology on the IBM Telco dataset yielded highly predictive results. The engineered financial volatility features allowed the CatBoost regressor to accurately model Risk-Adjusted Revenue, confirming the immense robustness of the original approach on public data.

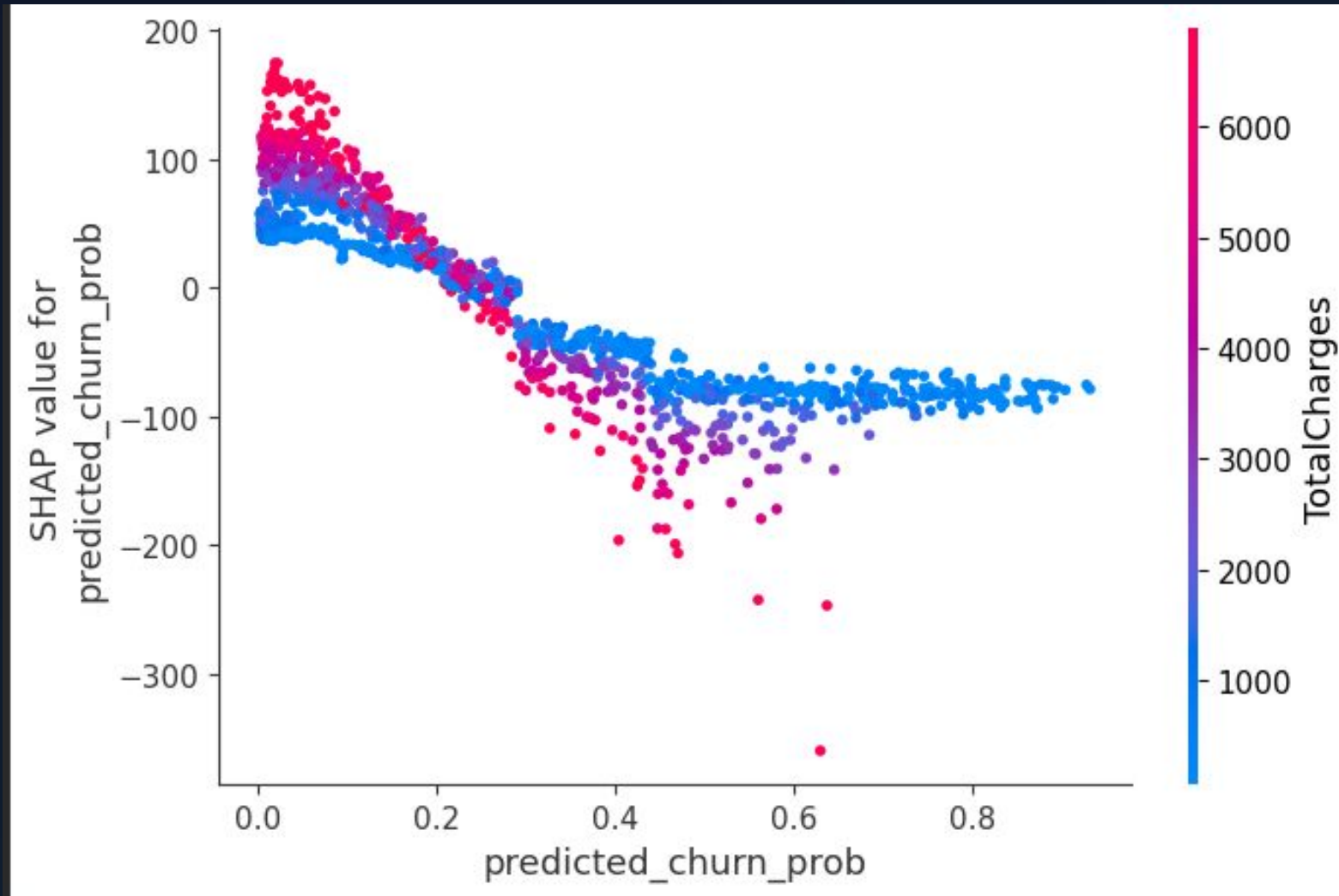
Explainability: SHAP Summary

Global Feature Importance

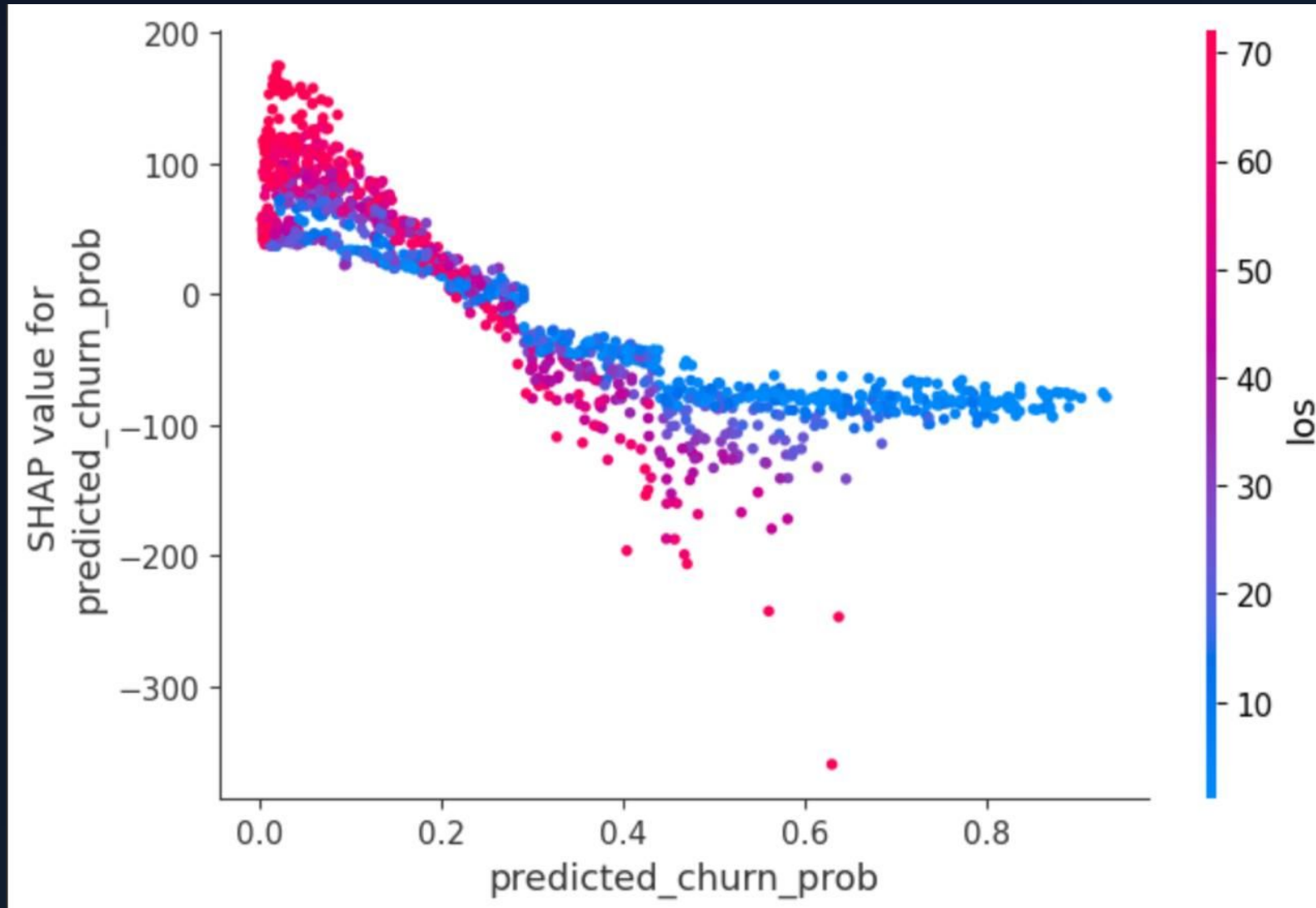
The summary plot confirms that financial volatility directly governs model predictions. High churn probabilities drastically penalize the risk-adjusted value of specific telecommunication customers.



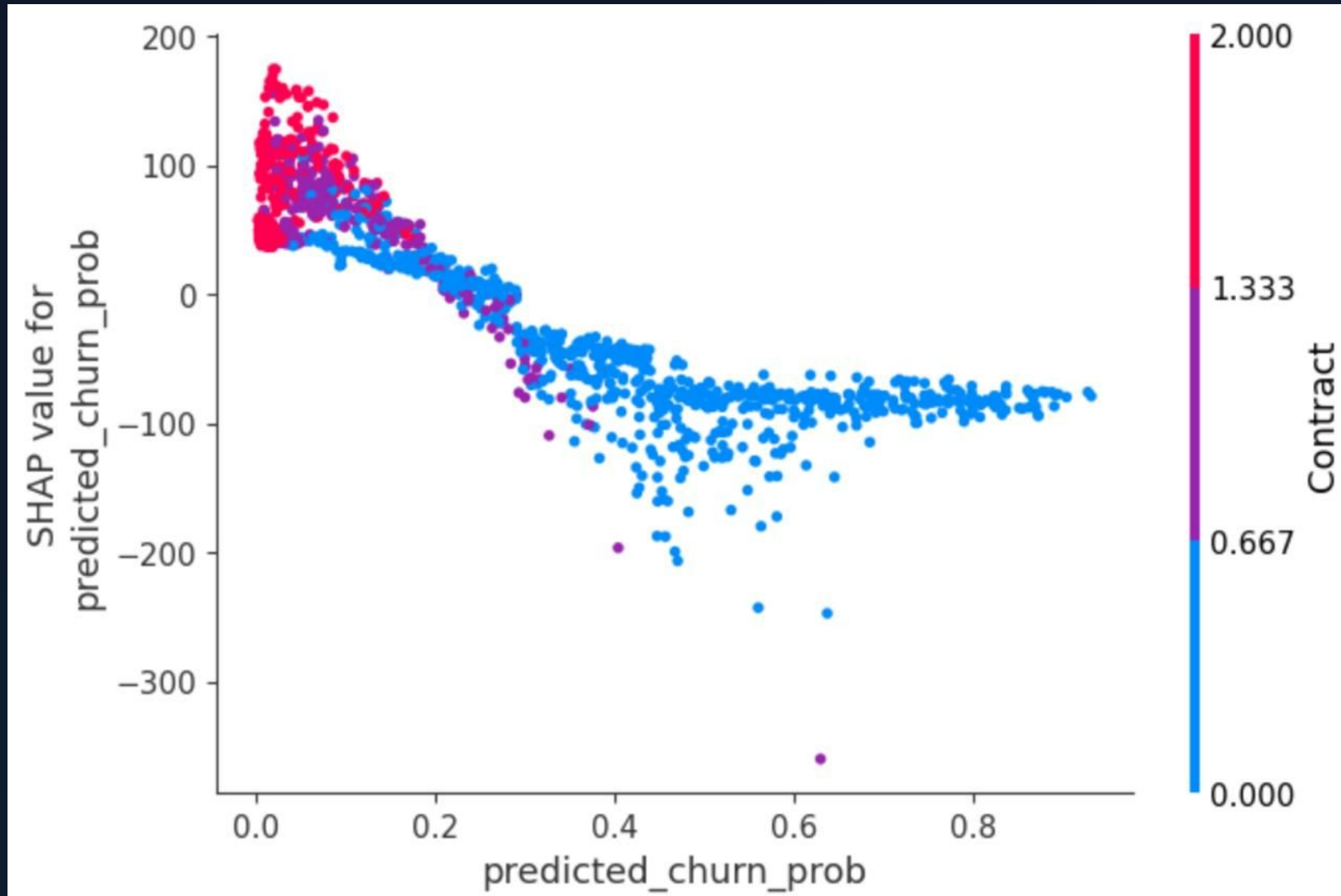
Demonstration of dependence between Churn and features



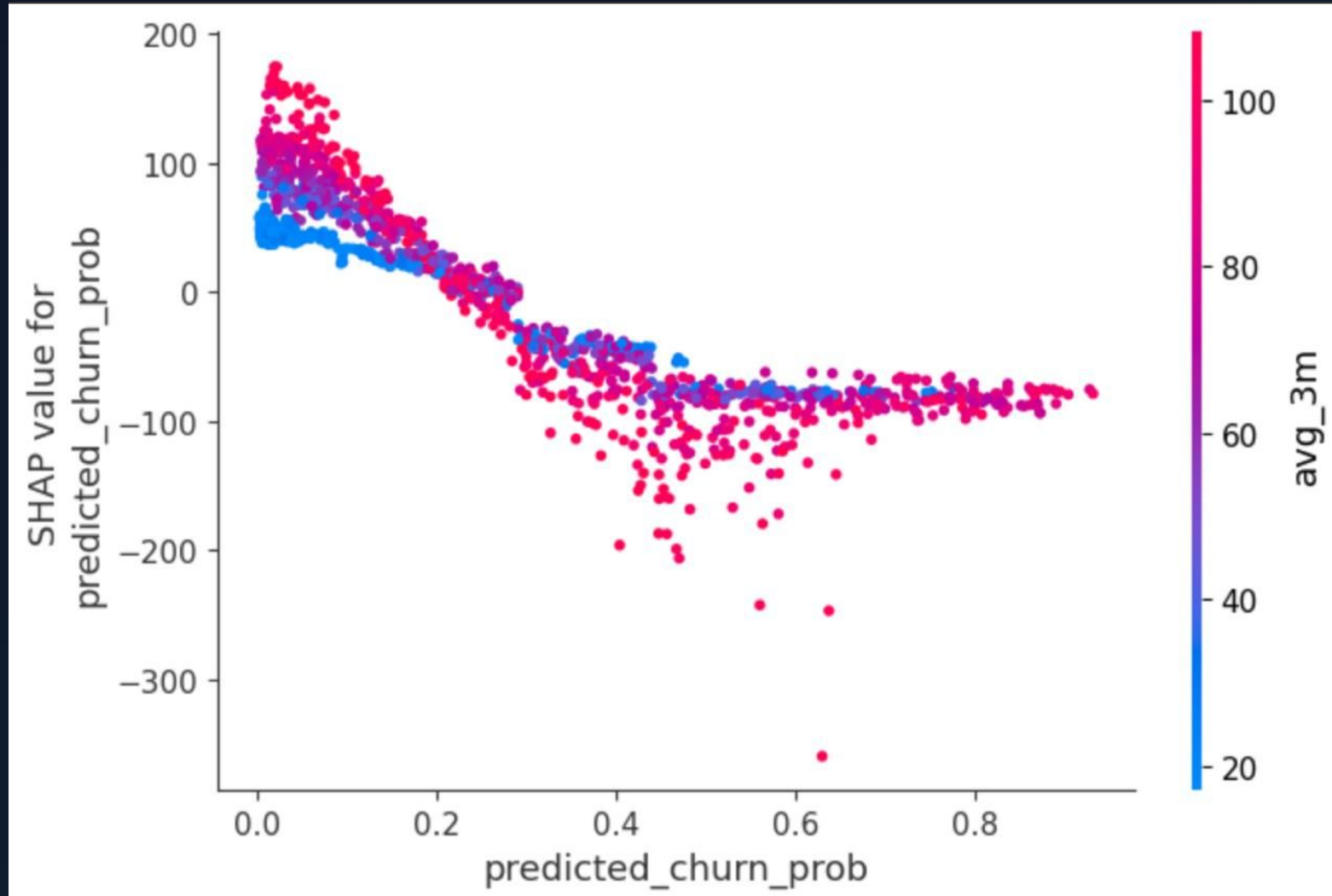
Demonstration of dependence between Churn and features



Demonstration of dependence between Churn and features



Demonstration of dependence between Churn and features



The Bottom Line

Telecommunication companies must urgently adopt **risk-adjusted valuation** models. Continuing to ignore individual churn probability and revenue volatility guarantees severe over-investment in high-risk customers, effectively destroying long-term portfolio value.

Thank you for your attention.